

titanic

May 2, 2025

```
[1]: print('this is Task5')
```

this is Task5

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[3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
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[5]: import scabon as sns
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ModuleNotFoundError                                Traceback (most recent call last)
Cell In[5], line 1
----> 1 import scabon as sns

ModuleNotFoundError: No module named 'scabon'
```

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[7]: import seaborn as sns
```

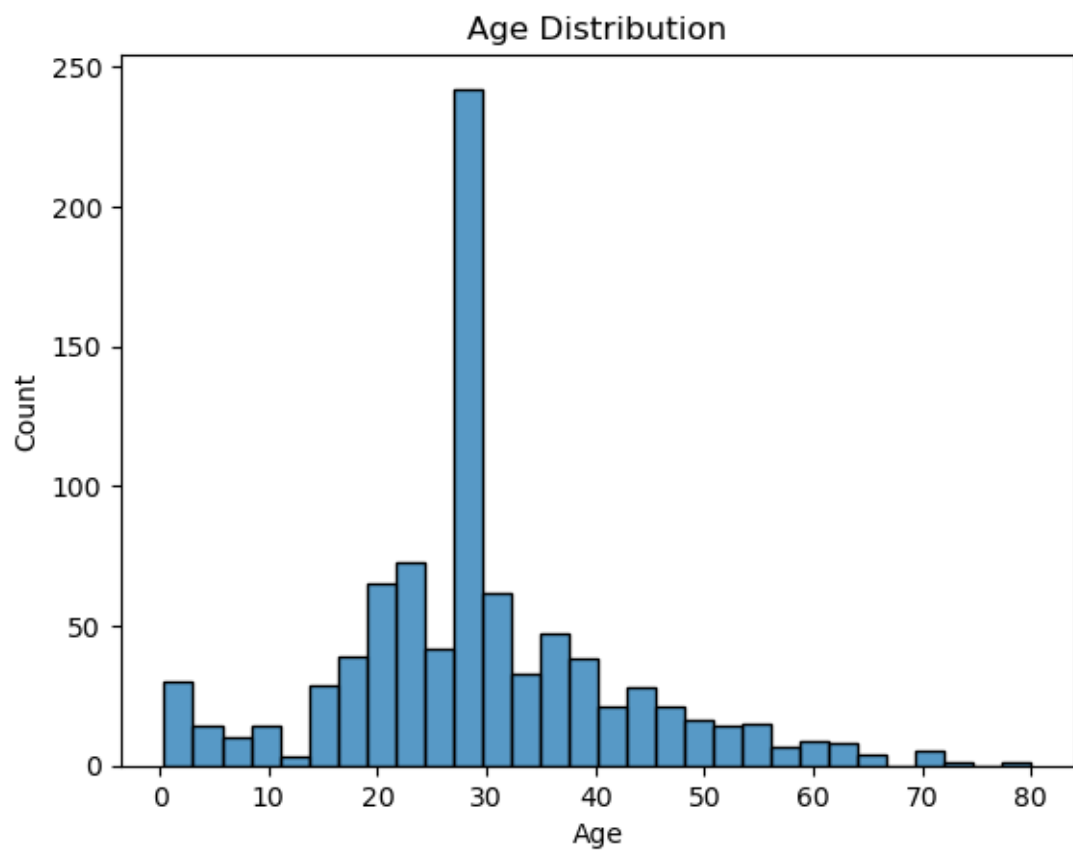
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[9]: df=pd.read_csv("titanic.csv")
```

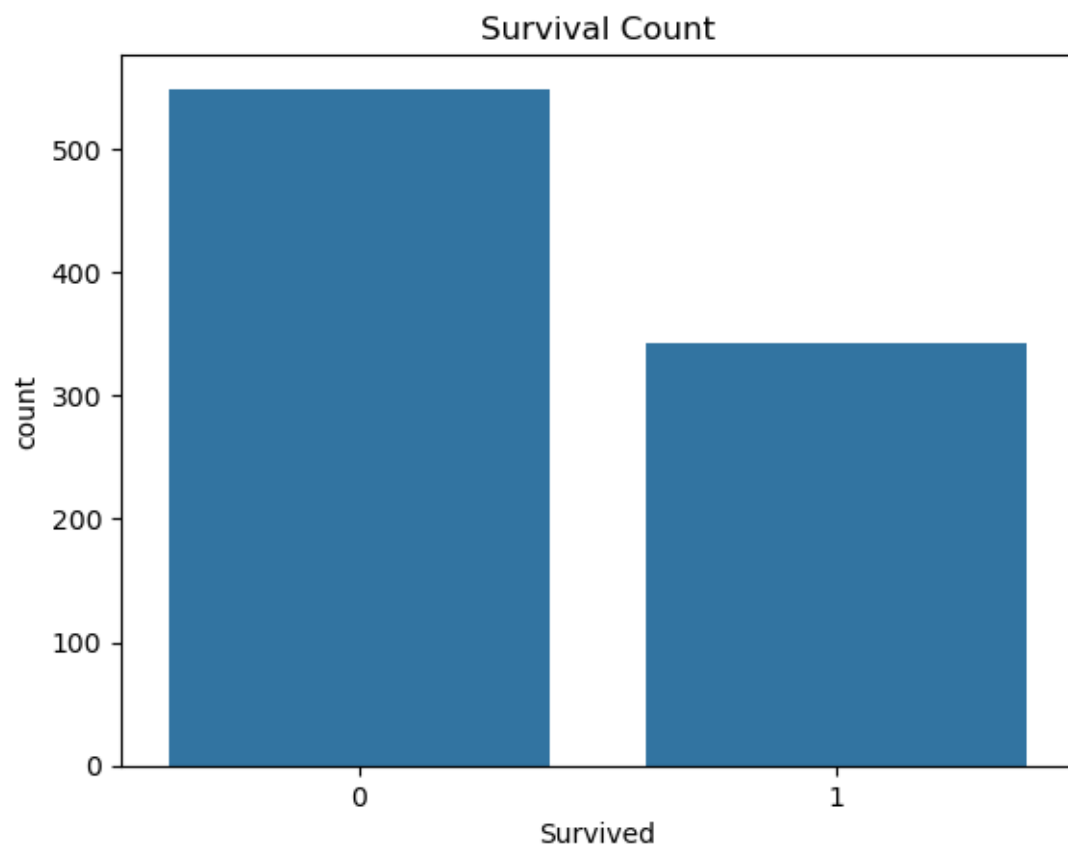
```
[11]: df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
if 'Cabin' in df.columns:
    df.drop(columns=['Cabin'], inplace=True)
```

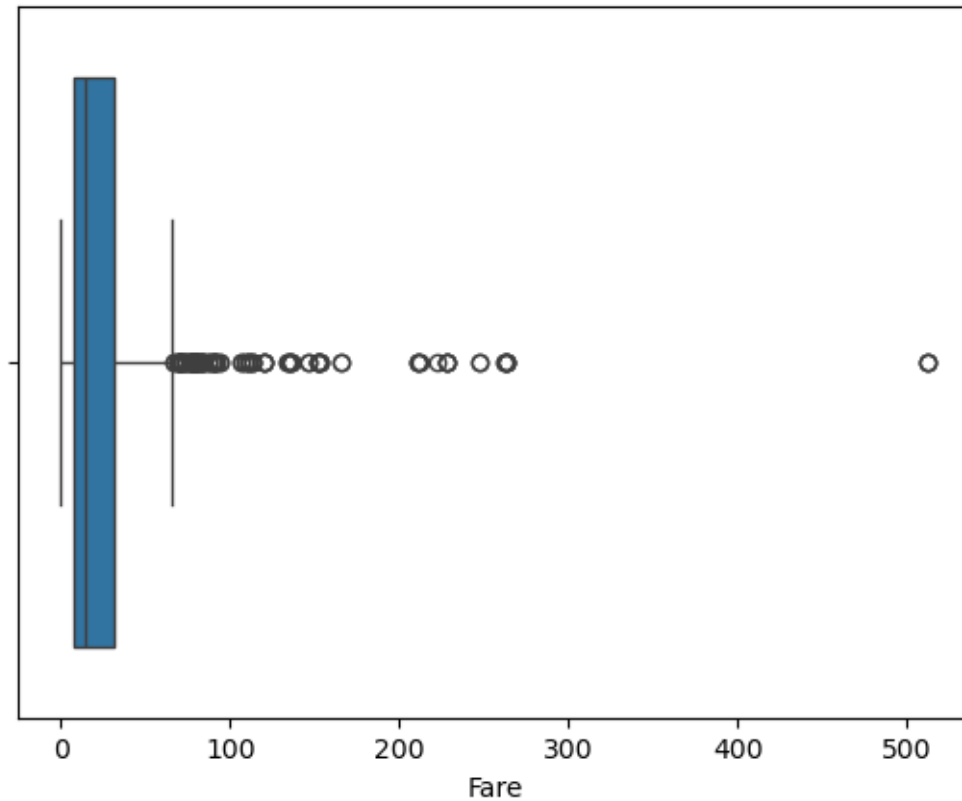
```
[13]: sns.histplot(df['Age'], bins=30)
plt.title("Age Distribution")
plt.show()

sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()

sns.boxplot(x=df['Fare'])
plt.show()
```



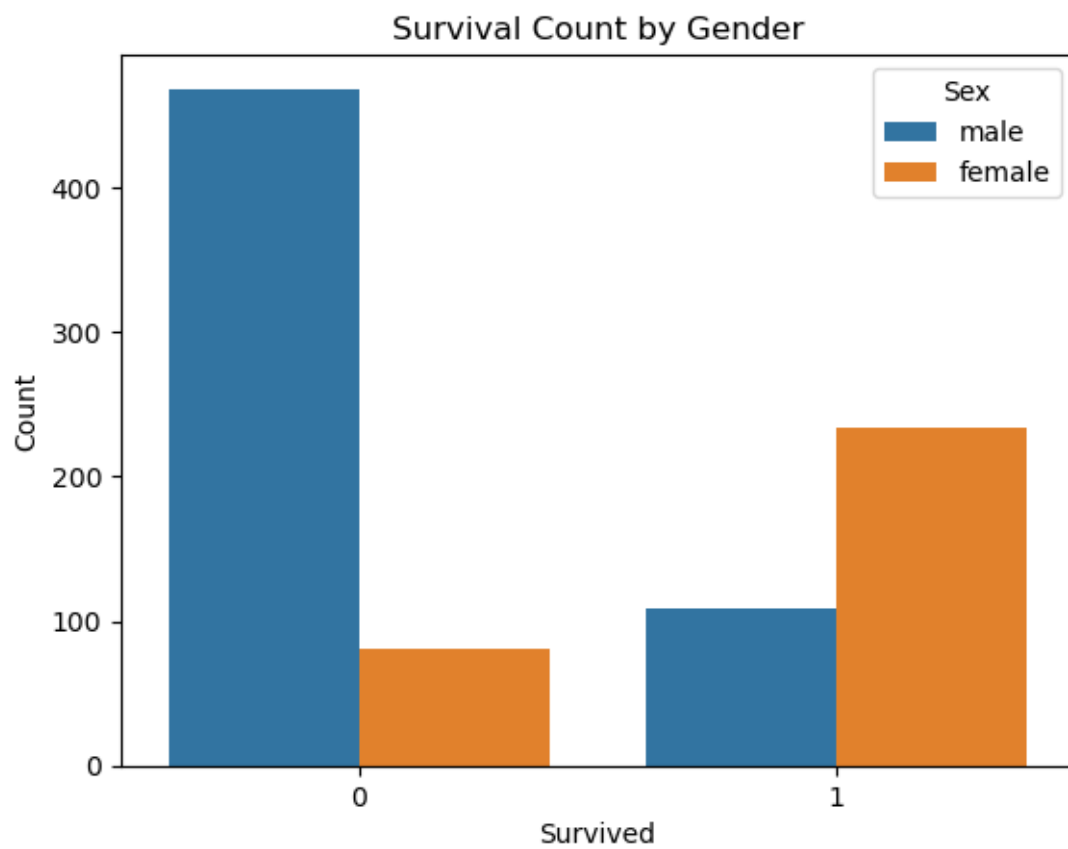


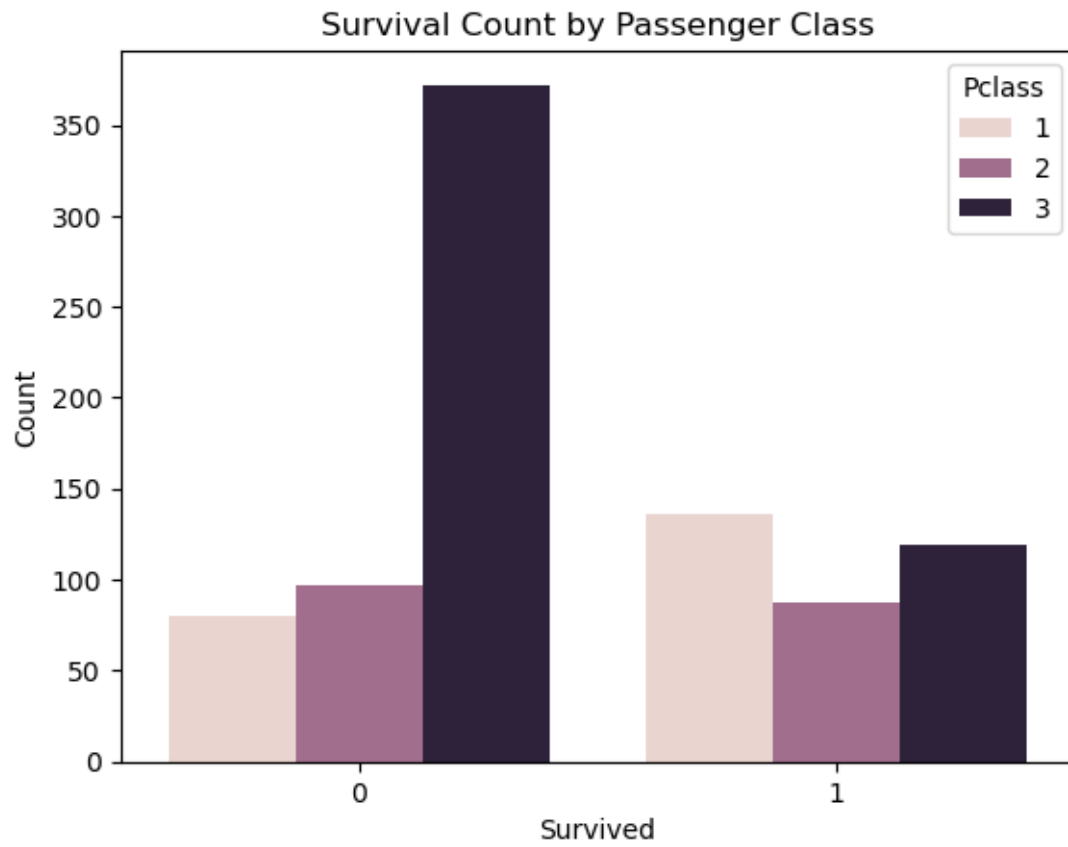


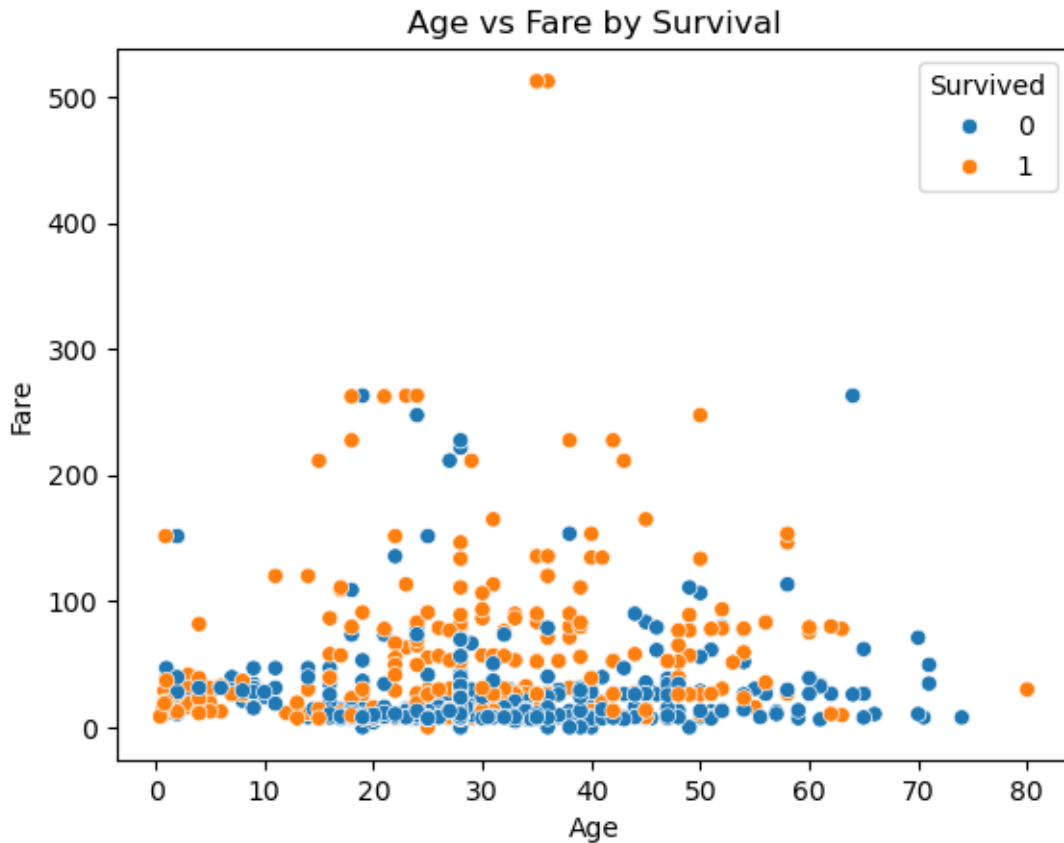
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[15]: # Survival by Sex
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title("Survival Count by Gender")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()

# Survival by Pclass
sns.countplot(x='Survived', hue='Pclass', data=df)
plt.title("Survival Count by Passenger Class")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()

# Age vs Fare
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

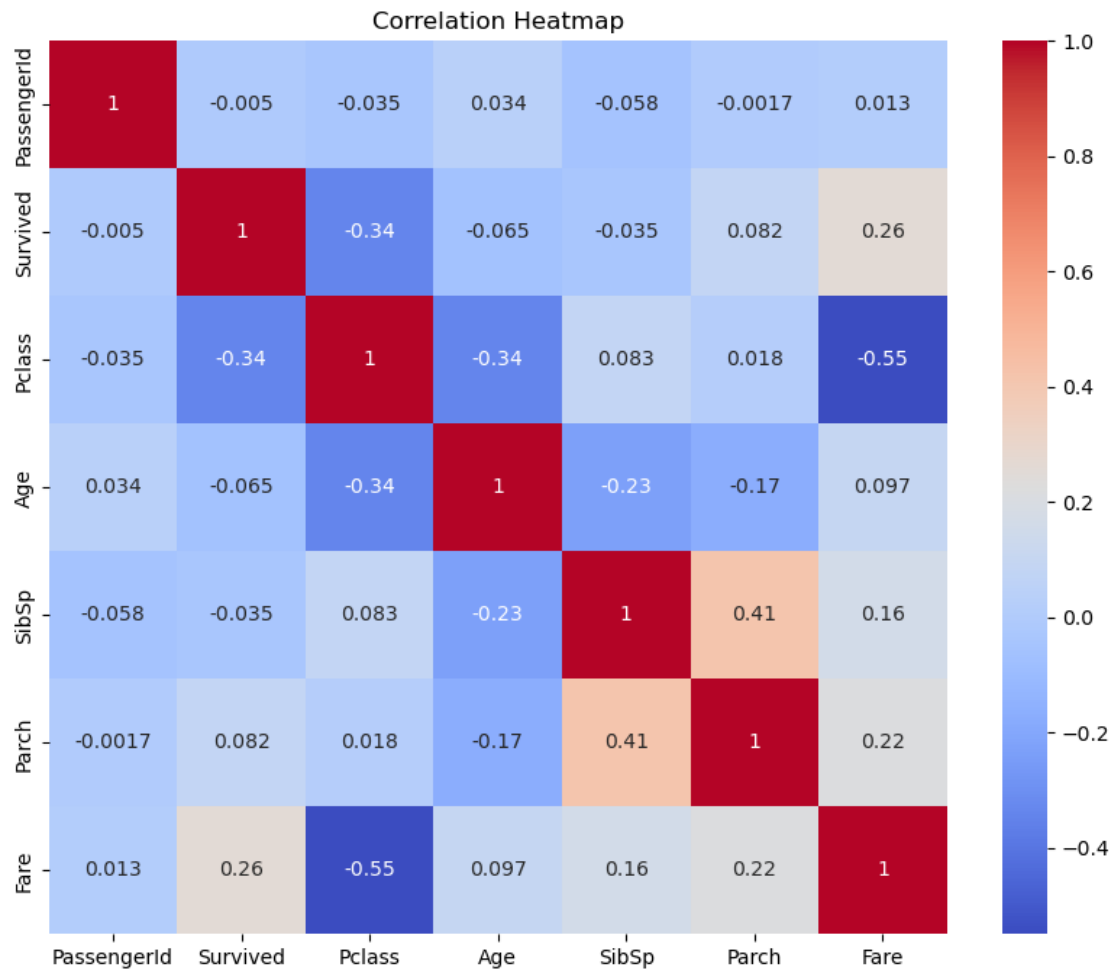


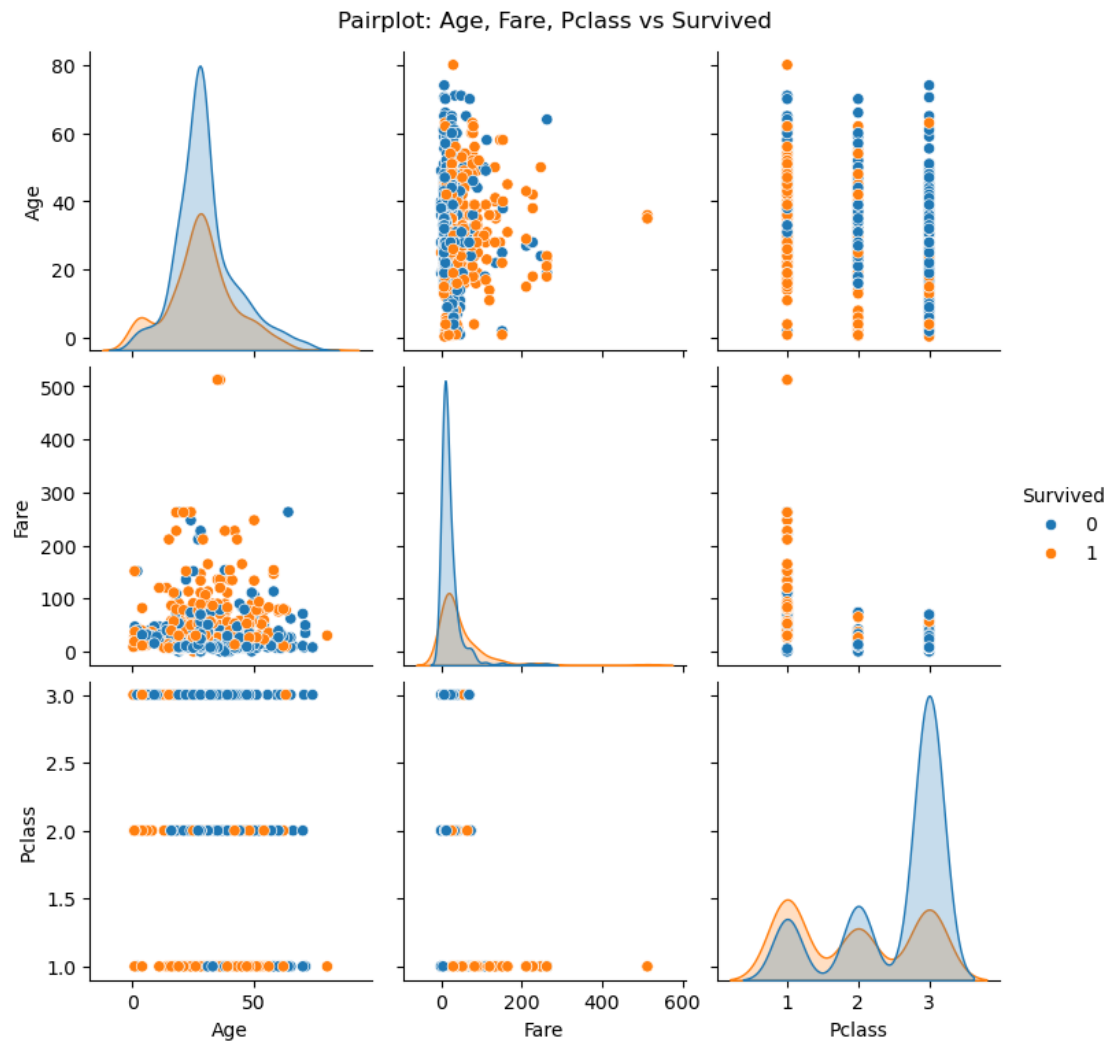




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[17]: # Heatmap of correlations
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# Pairplot of selected features
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']], hue='Survived')
plt.suptitle("Pairplot: Age, Fare, Pclass vs Survived", y=1.02)
plt.show()
```





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